

1. Say whether the limit exists and possibly find its value:

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{|x| + |y|}; \text{ Notice that } \left| \frac{xy}{|x| + |y|} \right| = \frac{|xy|}{|x| + |y|} = \begin{cases} \underbrace{\left[\frac{|x|}{|x| + |y|} \right]}_{\leq 1} |y| \leq \underbrace{|y|}_{\text{as } (x,y) \rightarrow (0,0)} \rightarrow 0. \\ \underbrace{\left[\frac{|y|}{|x| + |y|} \right]}_{\leq 1} |x| \leq \underbrace{|x|}_{\text{as } (x,y) \rightarrow (0,0)} \rightarrow 0. \end{cases}$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^4 y^2)}{x^4 + y^2}; \quad (\text{Hint: } \lim_{t \rightarrow 0} \frac{\sin t}{t} = ??).$$

$$\text{Notice that } \frac{\sin(x^4 y^2)}{x^4 + y^2} = \begin{cases} \frac{\sin(x^4 y^2)}{x^4 y^2} \times \frac{x^4 y^2}{x^4 + y^2} \rightarrow 0 \text{ as } (x, y) \rightarrow (0, 0) \\ \text{since } \frac{\sin(x^4 y^2)}{x^4 y^2} \rightarrow 1 \text{ from the hint and} \\ 0 \leq \frac{x^4 y^2}{x^4 + y^2} = \underbrace{\left[\frac{x^4}{x^4 + y^2} \right]}_{\leq 1} \times y^2 \leq y^2 \rightarrow 0 \end{cases}$$

We could also use $|\sin t| \leq |t|$ and get

$$0 \leq \frac{|\sin(x^4 y^2)|}{x^4 + y^2} \leq \frac{|x^4 y^2|}{x^4 + y^2} = \frac{x^4 y^2}{x^4 + y^2} = \begin{cases} \underbrace{\left[\frac{x^4}{x^4 + y^2} \right]}_{\leq 1} \times y^2 \leq y^2 \rightarrow 0 \\ \underbrace{\left[\frac{y^2}{x^4 + y^2} \right]}_{\leq 1} \times x^4 \leq x^4 \rightarrow 0. \end{cases}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{e^{(x^2+y^4)} - 1}{x^4 y^4}; \quad (\text{Hint: } \lim_{t \rightarrow 0} \frac{e^t - 1}{t} = ??).$$

$$\text{Using the hint, we get } \frac{e^{(x^2+y^4)} - 1}{x^4 y^4} \begin{cases} = \frac{e^{(x^2+y^4)} - 1}{x^2 + y^4} \times \frac{x^2 + y^4}{x^4 y^4}, \\ = \underbrace{\left[\frac{e^{(x^2+y^4)} - 1}{x^2 + y^4} \right]}_{\rightarrow 1 \text{ from the hint}} \times \underbrace{\left[\frac{1}{x^2 y^4} + \frac{1}{x^4} \right]}_{\rightarrow \infty} \end{cases}$$

Hence, $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{(x^2+y^4)} - 1}{x^4 y^4} = \infty$.

We could also use $e^{(x^2+y^4)} = e^{x^2} e^{y^4}$ and write

$$\frac{e^{(x^2+y^4)} - 1}{x^4 y^4} = \underbrace{\left[\frac{e^{x^2}}{x^4} \right]}_{\rightarrow \infty} \times \underbrace{\left[\frac{e^{y^4} - 1}{y^4} \right]}_{\rightarrow 1} \times \underbrace{\left[\frac{1}{x^2 y^4} \right]}_{\rightarrow \infty} \times \underbrace{\left[\frac{e^{x^2} - 1}{x^2} \right]}_{\rightarrow 1} \rightarrow \infty.$$

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{(x^2+y^4)} - 1}{|xy|}$; (Hint: $\lim_{t \rightarrow 0} \frac{e^t - 1}{t} = ??$).

Using the hint, we get $\frac{e^{(x^2+y^4)} - 1}{|xy|} = \underbrace{\left[\frac{e^{(x^2+y^4)} - 1}{x^2 + y^4} \right]}_{\rightarrow 1 \text{ from the hint}} \times \frac{x^2 + y^4}{|xy|}$

Now for the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^4}{|xy|}$ we find that

$$\begin{aligned} \frac{x^2 + y^4}{|xy|} &\stackrel{y=x}{=} \frac{x^2 + x^4}{x^2} = 1 + x^2 \rightarrow 1 \text{ as } (x, y) \rightarrow (0, 0), \\ &\stackrel{x=y^2}{=} \frac{y^4 + y^4}{|y|y^2} = 2|y| \rightarrow 0 \text{ as } (x, y) \rightarrow (0, 0). \end{aligned}$$

So, the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^4}{|xy|}$ does not exist and hence, the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{(x^2+y^4)} - 1}{|xy|}$ does not exist too.

(e) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy\sqrt{|xy|}}{x^2 + xy + y^2}$; (Hint: $x^2 + xy + y^2 \geq 0$; compare $|xy|$ and $x^2 + xy + y^2$).

Notice that

$$0 \leq \left| \frac{xy\sqrt{|xy|}}{x^2 + xy + y^2} \right| = \frac{|xy|\sqrt{|xy|}}{x^2 + xy + y^2} = \underbrace{\left[\frac{|xy|}{x^2 + xy + y^2} \right]}_{\leq 1 \text{ see } (*)} \times \sqrt{|xy|} \leq \underbrace{\sqrt{|xy|}}_{\text{as } (x,y) \rightarrow (0,0)} \rightarrow 0.$$

In fact,

$$|xy| \leq x^2 + xy + y^2 \text{ since } \begin{cases} xy \leq x^2 + xy + y^2 \\ -xy \leq x^2 + xy + y^2 \end{cases} \text{ and also } (*)$$

Therefore, $\frac{|xy|}{x^2 + xy + y^2} \leq 1$.

2. Let $\begin{cases} A := \{(x, y) \in \mathbb{R}^2: x^2 \leq y - 1\}, & B := \{(x, y) \in \mathbb{R}^2: y \leq 1\}, \\ C := \{(x, y) \in \mathbb{R}^2: 1 < y < x^2 + 1\}. \end{cases}$

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) := \begin{cases} 0 & \text{if } (x, y) \in A \cup B, \\ 1 & \text{if } (x, y) \in C, \end{cases}$ for every $(x, y) \in \mathbb{R}^2$.

- (i) Study the continuity of the function f at the point $(0, 1)$.

Notice that

$$\begin{cases} \lim_{\substack{(x,y) \rightarrow (0,1) \\ (x,y) \in A \cup B}} f(x,y) = \lim_{(x,y) \rightarrow (0,1)} 0 = 0, & (*) \\ \lim_{\substack{(x,y) \rightarrow (0,1) \\ (x,y) \in C}} f(x,y) = \lim_{(x,y) \rightarrow (0,1)} 1 = 1. & (**) \end{cases}$$

Thus, we get from $(*)$ and $(**)$ that the limit $\lim_{(x,y) \rightarrow (0,1)} f(x,y)$ does not exist and therefore, the function f is not continuous at $(0, 1)$.

Notice that, we could just evaluate the limit in $(**)$ and conclude that f is not continuous at $(0, 1)$ since $f(0, 1) = 0$.

- (ii) Find $f_x(0, 1)$ and $f_y(0, 1)$.

$$f_x(0, 1) = \lim_{x \rightarrow 0} \frac{f(x, 1) - f(0, 1)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0,$$

$$f_y(0, 1) = \lim_{y \rightarrow 0} \frac{f(0, 1 + y) - f(0, 1)}{y}.$$

To find $f(0, 1 + y)$ we need to consider the cases $y > 0$ and $y < 0$.

If $y > 0$, then $(0, 1 + y) \in A$ and hence,

$$f(0, 1 + y) = 0 \quad \Rightarrow \quad \lim_{y \rightarrow 0^+} \frac{f(0, 1 + y) - f(0, 1)}{y} = \lim_{y \rightarrow 0^+} \frac{0 - 0}{y} = 0.$$

On the other hand, if $y < 0$, then $(0, 1 + y) \in B$ and hence,

$$f(0, 1 + y) = 0 \quad \Rightarrow \quad \lim_{y \rightarrow 0^-} \frac{f(0, 1 + y) - f(0, 1)}{y} = \lim_{y \rightarrow 0^-} \frac{0 - 0}{y} = 0.$$

Therefore, we find that $f_y(0, 1) = 0$.

- (iii) Study the differentiability of the function f at the point $(0, 1)$.

The function is trivially not differentiable at the point $(0, 1)$ not being continuous at that point.

3. Let $\begin{cases} E := \{(x, y) \in \mathbb{R}^2: \alpha x \leq y \leq \beta x\} \text{ with } 0 < \alpha < \beta; \\ F := \{(x, y) \in \mathbb{R}^2: x > 0, y > 0\}. \end{cases}$

- (a) Explain why the domain E lies in the first quadrant, that $x \geq 0$ and $y \geq 0$.

If $(x, y) \in E$, then

$$\alpha x \leq y \leq \beta x \quad \Rightarrow \quad \alpha x \leq \beta x \quad \Rightarrow \quad (\beta - \alpha)x \geq 0 \quad \Rightarrow \quad x \geq 0 \quad \text{since } \beta > \alpha.$$

Thus, we get also $y \geq 0$ from $0 \leq \alpha x \leq y$. Therefore, the domain E is contained in the first quadrant.

- (b) Consider the function $f(x, y) := \frac{x^5 + y^5}{x^2 y^2}$.

Say whether the limits below exist and possibly find them:

(i) $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in E}} f(x,y)$. Notice that

$$\begin{aligned} (x,y) \in E &\Rightarrow \begin{cases} x^2 y^2 \geq \alpha^2 x^4 &\Rightarrow \frac{1}{x^2 y^2} \leq \frac{1}{\alpha^2 x^4} \\ x^5 + y^5 \leq x^5 + \beta^5 x^5 = (1 + \beta^5) x^5 \end{cases} \\ &\Rightarrow 0 \leq \frac{x^5 + y^5}{x^2 y^2} \leq \frac{(1 + \beta^5) x^5}{\alpha^2 x^4} = \frac{(1 + \beta^5) x}{\alpha^2} \longrightarrow 0. \end{aligned}$$

Thus, by the squeeze theorem, we find that $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in E}} f(x,y) = 0$.

(ii) $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in F}} f(x,y)$.

Hint: (ii) You can consider the parabolas $y = \alpha x^2$ with $0 < x < 1$ or $x = y^2$ with $0 < y < 1/\beta$.

Notice that $E \cap \text{dom}(f) \subset F$. So, on the one hand we obtained in (i) that

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in E}} f(x,y) = 0.$$

On the other hand, using the parabola $y = \alpha x^2$ with $0 < x < 1$, which is clearly contained in F and not in E , we get that

$$f(x, \alpha x^2) = \frac{x^5 + \alpha^5 x^{10}}{\alpha^2 x^6} = \frac{1}{\alpha^2 x} + \alpha^3 x^4 \longrightarrow \infty \text{ as } x \rightarrow 0^+.$$

So, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ y = \alpha x^2, \\ 0 < x < 1}} f(x,y) = \infty$. Therefore, the limit $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in F}} f(x,y)$ does not exist.

4. Let $G_1 := \{(x,y) \in \mathbb{R}^2: y \geq \lambda|x|, \lambda > 1\}$ and $G_2 := \{(x,y) \in \mathbb{R}^2: y > |x|\}$.

Consider the function $g(x,y) := \frac{x^2 y}{y^2 - x^2}$. Find the limits:

(a) $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in G_1}} g(x,y)$. Notice that

$$\begin{aligned} (x,y) \in G_1 &\Rightarrow y^2 \geq \lambda^2 x^2 \Rightarrow y^2 - x^2 \geq (1 - \lambda^2) x^2 \\ &\Rightarrow 0 \leq g(x,y) \leq \frac{x^2 y}{(\lambda^2 - 1) x^2} = \frac{y}{\lambda^2 - 1} \longrightarrow 0. \end{aligned}$$

Therefore, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in G_1}} g(x,y) = 0$ by the squeeze theorem.

$$(b) \quad \lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in G_2}} g(x,y).$$

Hint: (b) You can consider for instance the arc of curve $y = |x| + x^2$ for $|x| > 0$ and definitely small. Then, use (a) to conclude.

Notice that $G_1 \subset G_2$ and we got in (a) that $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in G_1}} g(x,y) = 0$.

On the other hand, using the arc of curve $y = |x| + x^2$ which is contained in G_2 and not in G_1 for $|x|$ small, we obtain that

$$\begin{aligned} g(x, |x| + x^2) &= \frac{x^2(|x| + x^2)}{(|x| + x^2)^2 - x^2} = \frac{x^2(|x| + x^2)}{\underbrace{(|x|^2 + 2|x|x^2 + x^4) - x^2}_{|x|^2 = x^2}} = \frac{x^2(|x| + x^2)}{(x^2 + 2|x|x^2 + x^4) - x^2} \\ &= \frac{x^2(|x| + x^2)}{x^2(2|x| + x^2)} = \frac{|x| + x^2}{2|x| + x^2} = \frac{|x|(1 + |x|)}{\underbrace{|x|(2 + |x|)}_{|x|^2 = x^2}} = \frac{1 + |x|}{2 + |x|} \longrightarrow \frac{1}{2}. \end{aligned}$$

Hence, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ y = |x| + x^2, \\ |x| \text{ small}}} g(x,y) = \frac{1}{2}$ which together with (a), show that $\lim_{\substack{(x,y) \rightarrow (0,0) \\ (x,y) \in G_2}} g(x,y)$ does not exist.

5. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x,y) := \begin{cases} \frac{x^3 - y^3}{2x^2 + xy + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$

(i) Study the continuity of the function f at the point $(0,0)$.

(Hint: you can use for instance $1 + \cos^2(\theta) + \sin \theta \cos \theta = 1 + \cos^2(\theta) + \frac{1}{2} \sin(2\theta) \geq 0$).

We want to check whether $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0) = 0$.

First solution: We find along the axes and along the line $y = x$ that

$$f(x, 0) = \frac{x^3}{2x^2} = \frac{x}{2} \rightarrow 0, \quad f(0, y) = \frac{-y^3}{y^2} = -y \rightarrow 0 \text{ while } f(x, x) = 0.$$

So, we get the candidate $L = 0$.

Notice that $2x^2 + xy + y^2 = x^2 + (x^2 + xy + y^2) = x^2 + \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2 \geq 0$.

Now, we use polar coordinates r, θ through $x = r \cos \theta$ and $y = r \sin \theta$ and get

$$\begin{aligned}
0 \leq |f(r \cos \theta, r \sin \theta) - 0| &= \frac{|r^3 \cos^3 \theta - r^3 \sin^3 \theta|}{2r^2 \cos^2 \theta + r^2 \cos \theta \sin \theta + r^2 \sin^2 \theta} \\
&= \frac{r |\cos^3 \theta - \sin^3 \theta|}{\cos^2 \theta + 1 + \frac{1}{2} \sin(2\theta)} \leq \frac{r(|\cos^3 \theta| + |\sin^3 \theta|)}{1 + \frac{1}{2} \sin(2\theta)} \\
&\leq \frac{2r}{\frac{1}{2}} = 4r \rightarrow 0 \text{ as } r \rightarrow 0^+.
\end{aligned}$$

where we have used

$$\begin{aligned}
2 \cos^2 \theta + \cos \theta \sin \theta + \sin^2 \theta &= \cos^2 \theta + \cos \theta \sin \theta + (\cos^2 \theta + \sin^2 \theta) \\
&= \cos^2 \theta + 1 + \frac{1}{2} \sin(2\theta) \geq 1 + \frac{1}{2} \sin(2\theta) \geq 1 - \frac{1}{2} = \frac{1}{2}.
\end{aligned}$$

Hence, $\lim_{r \rightarrow 0^+} f(r \cos \theta, r \sin \theta) = 0$ uniformly in θ and therefore, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$ and thus, the function f is continuous at $(0, 0)$.

Second solution: We find along the axes and along the line $y = x$ that the limit is equal to 0 as above. Now using $2x^2 + xy + y^2 = x^2 + (x + \frac{y}{2})^2 + \frac{3}{4}y^2$ we get

$$\begin{aligned}
0 \leq |f(x, y) - 0| &= \frac{|x^3 - y^3|}{x^2 + (x + \frac{y}{2})^2 + \frac{3}{4}y^2} \leq \frac{|x^3| + |y^3|}{x^2 + (x + \frac{y}{2})^2 + \frac{3}{4}y^2} \\
&= \frac{|x^3|}{x^2 + (x + \frac{y}{2})^2 + \frac{3}{4}y^2} + \frac{|y^3|}{x^2 + (x + \frac{y}{2})^2 + \frac{3}{4}y^2} \\
&\leq \frac{|x^3|}{x^2} + \frac{|y^3|}{\frac{3}{4}y^2} = \underbrace{|x| + \frac{4}{3}|y|}_{\text{as } (x,y) \rightarrow (0,0)} \rightarrow 0.
\end{aligned}$$

Hence, we get by the squeeze theorem that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$ and thus, the function f is continuous at $(0, 0)$.

(ii) Find $f_x(0, 0)$ and $f_y(0, 0)$.

$$\begin{cases} f_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{x - 0}{2x} = \frac{1}{2} \\ f_y(0, 0) = \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{-y - 0}{y} = -1. \end{cases}$$

(iii) Study the differentiability of the function f at the point $(0, 0)$.

Hint: Use the definition of differentiability and check the limit along the line $y = x$.

We want to check whether $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0) - f_x(0, 0)(x - 0) - f_y(0, 0)(y - 0)}{\sqrt{(x - 0)^2 + (y - 0)^2}} = 0$.

Notice that

$$\begin{aligned}
& \frac{f(x, y) - f(0, 0) - f_x(0, 0)(x - 0) - f_y(0, 0)(y - 0)}{\sqrt{(x - 0)^2 + (y - 0)^2}} = \frac{\frac{x^3 - y^3}{2x^2 + xy + y^2} - \frac{1}{2}x + y}{\sqrt{x^2 + y^2}} \\
& = \frac{x^3 - y^3 + \left(-\frac{1}{2}x + y\right)(2x^2 + xy + y^2)}{(2x^2 + xy + y^2)\sqrt{x^2 + y^2}} = \frac{x^3 - y^3 - x^3 - \frac{1}{2}x^2y - \frac{1}{2}xy^2 + 2x^2y + xy^2 + y^3}{(2x^2 + xy + y^2)\sqrt{x^2 + y^2}} \\
& = \frac{\frac{3}{2}x^2y + \frac{1}{2}xy^2}{(2x^2 + xy + y^2)\sqrt{x^2 + y^2}} \quad \xrightarrow{\text{as } (x, y) \rightarrow (0, 0)} \left\{ \begin{array}{ll} 0 & \text{along both axes} \\ \frac{1}{2\sqrt{2}} & \text{along } y = x \text{ with } x > 0; \\ -\frac{1}{2\sqrt{2}} & \text{along } y = x \text{ with } x < 0. \end{array} \right.
\end{aligned}$$

Therefore, the function f is not differentiable at the point $(0, 0)$.